

Community Engagement for Prevention of Communicable Diseases

FINAL TERM PROJECT REPORT

HealthGuard UOL

Course: Civic & Community Engagement

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Abstract

Communicable diseases continue to pose a significant public health challenge in Pakistan, contributing to morbidity, mortality, and socioeconomic burden. Despite improvements in healthcare infrastructure, disease transmission persists due to inadequate hygiene practices, low public awareness, and insufficient preventive measures in community spaces.

The HealthGuard UOL project is a student-led community engagement initiative aimed at preventing communicable diseases through education, visual awareness campaigns, and distribution of personal protective equipment (PPE). Interventions were implemented across **21 businesses** including gyms, cafes, retail stores, and barber salons in Lahore, Sheikhpura, and Swat. A total of **51 surveys** were collected to establish baseline conditions.

To assess long-term effectiveness, a **follow-up visit on 15th December 2025** confirmed that **80 awareness posters were still displayed** and approximately **80% of businesses** continued compliance with hygiene guidelines, indicating sustained behavioral change.

Aligned with **Sustainable Development Goal 3 (Good Health and Well-Being)**, the project demonstrates that low-cost, student-led interventions can generate measurable and lasting improvements in community health.

1. Introduction

Communicable diseases, caused by bacteria, viruses, fungi, and parasites, remain a critical health concern in Pakistan. These illnesses spread through direct contact, respiratory droplets, contaminated surfaces, food, water, or vectors. Overcrowding, poor sanitation, and limited health literacy exacerbate the risks, particularly in gyms, cafes, retail stores, and barber salons.

Despite government-led programs, these high-traffic community spaces are often neglected in preventive strategies. HealthGuard UOL aims to address this gap by emphasizing **community engagement** as a cornerstone of public health, ensuring that disease prevention responsibilities are shared among healthcare providers, governments, citizens, and business owners.

2. Problem Identification and Baseline Situation

2.1 National Communicable Disease Burden

Communicable diseases constitute approximately **38% of Pakistan's total disease burden**, with tuberculosis, malaria, dengue, typhoid, diarrheal diseases, and influenza-like illnesses being most prevalent. Approximately 510,000 new TB cases occur annually, and dengue outbreaks recur seasonally.

2.2 Baseline Findings

Baseline observations and surveys highlighted:

- Limited availability and use of hand sanitizers
- Rare adoption of PPE
- Lack of health awareness posters
- Low perceived importance of preventive practices

These trends demonstrated the urgent need for targeted interventions in community spaces.

3. Literature Review and Theoretical Framework

The project draws on global and national health strategies. WHO emphasizes hygiene and prevention as primary defense mechanisms. Community engagement theory supports active participation as a key to sustainable behavior change. Behavior Change Communication (BCC) highlights visual cues, repeated messaging, and interpersonal engagement to reinforce preventive practices.

4. Literature Summary Table

Source	Year	Key Idea	Relevance to Project
WHO – Communicable Disease Prevention	2023	Global prevention strategies	Guided intervention design
United Nations – SDG 3	2015	Universal health coverage	SDG alignment
NIH Pakistan – IDSR Reports	2022	Disease surveillance	Local context
Punjab Health Dept. – Disease Control	2021	Provincial programs	Guided local intervention
WHO – Hand Hygiene Guidelines	2009	Handwashing best practices	Basis for hygiene education
BMC Public Health – Community Interventions	2020	Community engagement effectiveness	Methodological support
Healthwire Pakistan – Communicable Diseases	2023	Current disease trends	Local relevance
Joint SDG Fund – Health & Community Engagement	2022	Integrated approaches	Sustainability & replication

Each source informed the design and execution of HealthGuard UOL, ensuring evidence-based, sustainable, and scalable interventions.

5. Research Methodology

A **mixed-methods design** was employed, combining quantitative surveys (51 respondents) and qualitative observations across **21 businesses** in Lahore, Sheikhpura, and Swat. Ethical considerations included voluntary participation, informed consent, and confidentiality. This approach enabled comprehensive understanding of community health behaviours.

6. Project Design and Action Plan

Phased implementation included:

1. Baseline research and needs assessment
2. Resource procurement and awareness material creation
3. On-site community engagement and PPE/poster distribution
4. Documentation and monitoring
5. Follow-up evaluation

Team roles: Asad Ishaq – social media & reporting , Osman – survey coordination, Shaharyar & Umer – business engagement, Abdullah – documentation.

7. Community Engagement Activities

Targeted interventions were implemented in:

- **Gyms:** Equipment sanitization, PPE education
- **Cafes/Restaurants:** Food hygiene
- **Retail Stores:** Awareness posters, customer engagement
- **Barber Salons:** Tool sterilization, PPE use

Each site was selected for high interaction and transmission risk.

8. Social Media Campaign

Instagram handle **@healthguard_uol** reached over 10,000+ views with 250+ engagements. Hashtags included #HealthGuardUOL, #PreventDisease, #CommunityHealth, #SDG3. Digital advocacy amplified on-ground interventions and extended reach.

9. Post-Intervention Impact and Trend Analysis

Observed improvements included:

- Increased sanitizer availability and usage
- Higher PPE compliance
- Posters visible in all locations
- Enhanced customer participation

Trends show measurable positive shifts in hygiene behavior from baseline.

10. Follow-Up Visit and Sustainability Assessment

Date: 15 December 2025

Key Findings:

- 80 awareness posters still displayed
- ~80% of businesses complying with hygiene guidelines
- Continued sanitizer and PPE usage
- Regular cleaning routines maintained

This confirms sustained behavioral change and effectiveness of combining education, visual cues, and community engagement.

11. Sustainability and Scalability

Short-term: Regular follow-ups and poster replenishment

Medium-term: Partnerships with medical students, expansion to 15-20 more businesses

Long-term: Institutionalization via student societies, government collaboration, certification programs, mobile apps, and training videos

Low-cost and scalable model allows replication across multiple areas.

12. Innovation

- Posters as behavioral nudges
- Social media amplification
- Student-led leadership model
- Data-driven monitoring and reporting

13. Challenges and Limitations

- Resistance from some businesses
- Photography/video restrictions
- Time constraints for follow-ups
- Budget limitations
- Coordination across 5 team members

14. Lessons Learned and Ethical Reflection

- Trust-building is crucial
- Low-cost interventions can create lasting change
- Visual aids are more effective than verbal instructions alone
- Partnering with business owners as allies increases sustainability
- Proper documentation strengthens credibility

15. SDG Alignment and Policy Implications

- **SDG 3:** Promotes health and well-being
- **SDG 11:** Contributes to safe, inclusive communities
- **SDG 17:** Encourages partnerships for sustainable development

Project serves as a model for student-led, scalable interventions aligned with national and global health priorities.

16. Conclusion

HealthGuard UOL demonstrates that student-led community engagement can produce measurable, sustainable improvements in public health. Follow-up visits confirmed the lasting impact of awareness campaigns and hygiene interventions. Combining education, visual reminders, and stakeholder collaboration promotes behavioral change and long-term community health.

17. References

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